

Cambridge (CIE) IGCSE Additional Maths



Your notes

Vectors in Two Dimensions

Contents

- * Introduction to Vectors
- * Vector Addition
- * Problem Solving using Vectors
- * Compose & Resolve Velocities



Your notes

Introduction to Vectors

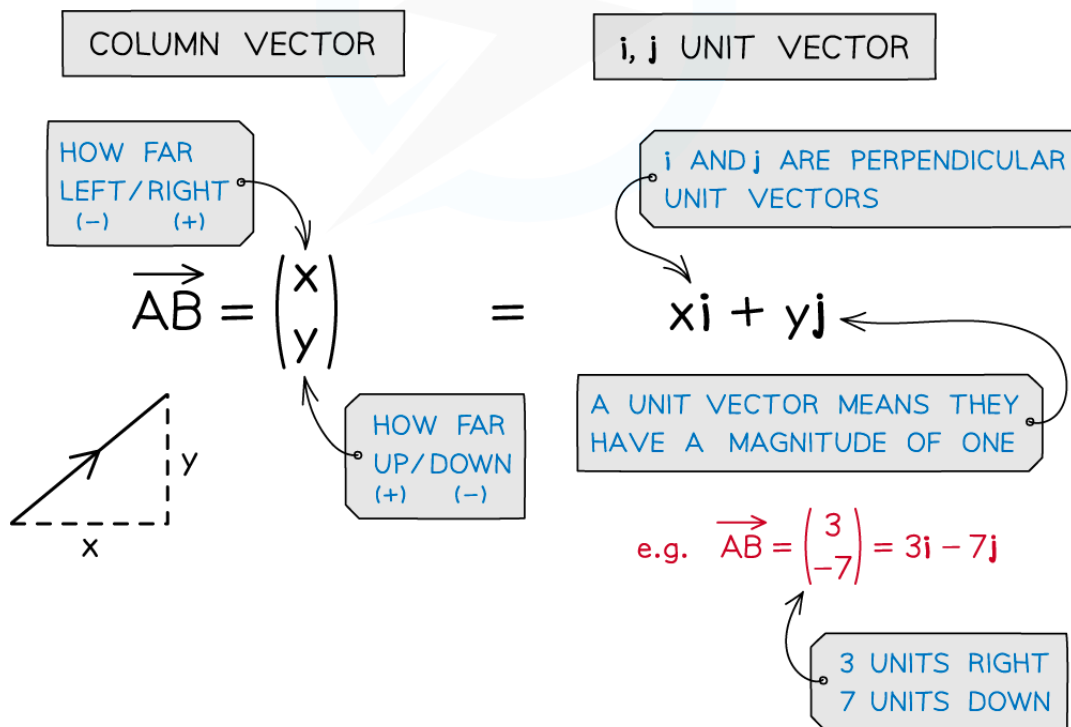
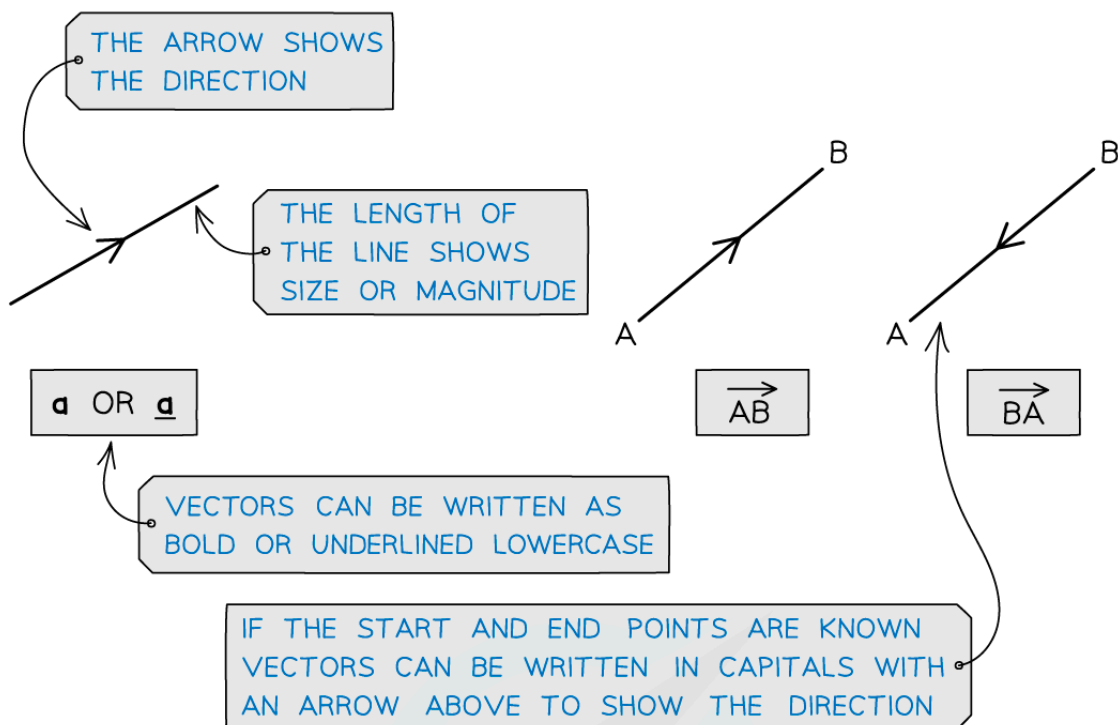
Basic Vectors

What is a vector?

- Vectors represent a movement of a certain **magnitude** (size) in a given **direction**
 - For example: two objects with **velocities** of 7 m/s and -7 m/s are travelling at the **same speed** but in **opposite directions**
- You should have already come across **vectors** when translating functions of graphs
- They appear in many contexts of maths including **mechanics** for modelling forces
- A vector in two directions has **components** in the direction of the x- and y- axes
 - Vector quantities can have **positive** or **negative** components
- Vectors can be represented in different ways such as a **column vector** or as an **i** and **j** unit vector
- Some examples of vector quantities you may come across are **displacement**, **velocity** or **acceleration**



Your notes





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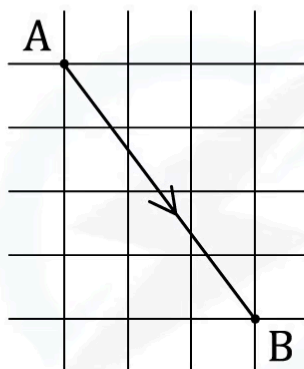
Examiner Tips and Tricks

- Think of vectors like a journey from one place to another
- Diagrams can help, if there isn't one, draw one
- In your exam you can't write in **bold** so should underline your vector notation



Worked Example

- ? a) From the diagram below, write the column vector for $\vec{AB}$.



- b) Write the column vector $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$ using the unit vectors $\mathbf{i}$ and $\mathbf{j}$.

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Your notes

a) $\vec{AB} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$

REMEMBER THE COLUMN VECTOR SIMPLY REPRESENTS THE PATH (LEFT/RIGHT UP/DOWN) FROM A TO B

b) $\begin{pmatrix} -2 \\ 5 \end{pmatrix} = -2\mathbf{i} + 5\mathbf{j}$

THE VECTOR USING $\mathbf{i}$ AND $\mathbf{j}$

IN YOUR EXAM YOU WOULD WRITE $-2\mathbf{i} + 5\mathbf{j}$

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Magnitude of a Vector

How do you find the magnitude of a vector?

- The **magnitude** of a vector tells us its **size** or **length**
- The magnitude of the vector $\vec{AB}$ is denoted $|\vec{AB}|$
 - The magnitude of the vector $\mathbf{a}$ is denoted $|\mathbf{a}|$
- The magnitude of a vector can be found using Pythagoras' theorem
- The magnitude of a vector $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j}$ is found using

- $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2}$

- where $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$

MAGNITUDE

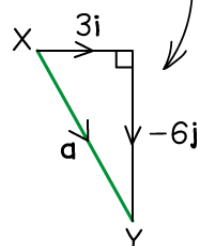
$$|a| = |xi + yj| = \left| \begin{pmatrix} x \\ y \end{pmatrix} \right| = \sqrt{x^2 + y^2}$$

REMEMBER THERE ARE LOTS OF DIFFERENT WAYS TO REPRESENT THE SAME VECTOR

MAGNITUDE

$$|a| = |\vec{XY}| = \left| \begin{pmatrix} 3 \\ -6 \end{pmatrix} \right| = |3i - 6j|$$

VECTORS MAGNITUDE IS SOMETIMES REFERRED TO AS ITS MODULUS



$$|\vec{XY}| = \sqrt{3^2 + 6^2} = 3\sqrt{5} = 6.708... = 6.7 \text{ (TO 1 dp)}$$

YOU CAN IGNORE MINUS SIGN

IF ASKED FOR THE EXACT VALUE LEAVE YOUR ANSWER IN SIMPLIFIED SURD FORM

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What is a unit vector?

- A **unit vector** has a **magnitude** of 1
- The vectors **i** and **j** are unit vectors
 - the direction of **i** is in the positive **x**-direction
 - the direction of **j** is in the positive **y**-direction
- To find a unit vector in the direction of a given vector **divide** the **vector** by its **magnitude**





Your notes

Examiner Tips and Tricks

- Finding the magnitude of a vector is the same as finding the distance between two coordinates
- Commit the formula to memory and be prepared to use it in the exam



Worked Example

A vector $\vec{XY} = \begin{pmatrix} -12 \\ 5 \end{pmatrix}$.

a) Find $|\vec{XY}|$.

$|\vec{XY}|$ is the magnitude of the vector, so use Pythagoras' theorem.

$$|\vec{XY}| = \sqrt{12^2 + 5^2} = \sqrt{169}$$

$$|\vec{XY}| = 13$$

b) Find the unit vector in the direction of $\vec{XY}$.

For a unit vector, divide the vector $\vec{XY}$ by its magnitude, which was found in part (a).

$$\frac{\vec{XY}}{|\vec{XY}|} = \frac{-12\mathbf{i} + 5\mathbf{j}}{13} = -\frac{12}{13}\mathbf{i} + \frac{5}{13}\mathbf{j}$$

The unit vector in the direction of $\vec{XY}$ is $\begin{pmatrix} -12/13 \\ 5/13 \end{pmatrix}$.

Position Vectors

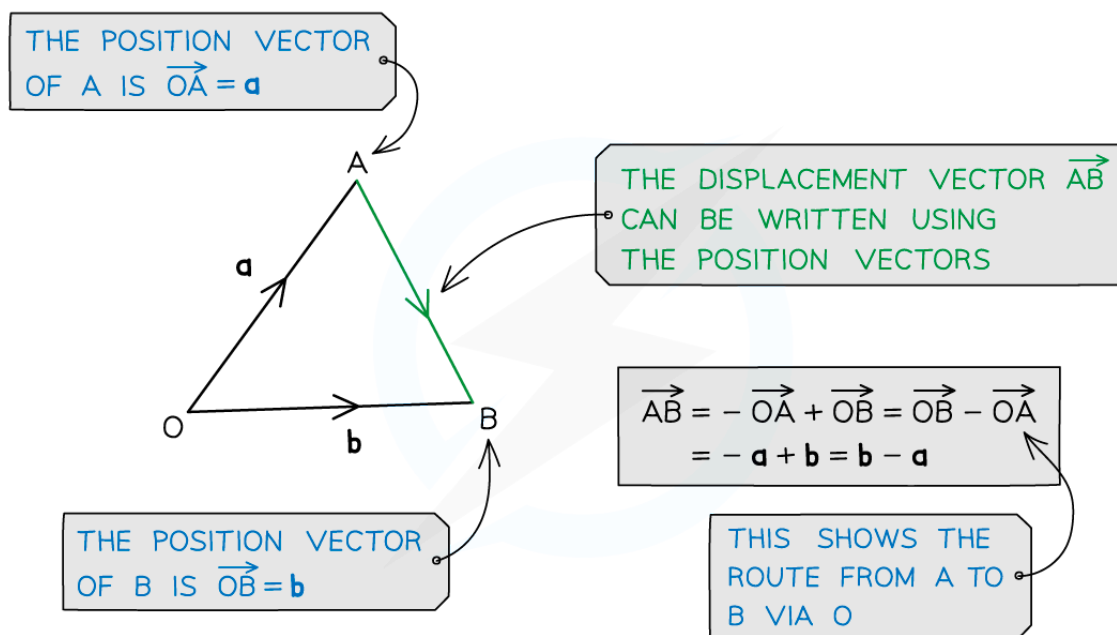
What is a position vector?

- Position vectors describe the **position of a point** in relation to the origin
- They are different to **displacement vectors** which describe the **direction and distance** between any two points



Your notes

- The position vector of point A is written with the notation $\mathbf{a} = \overrightarrow{OA}$
 - The origin is always denoted O
- The individual components of a position vector are the coordinates of its end point
 - For example the point with coordinates (3, -2) has position vector $3\mathbf{i} - 2\mathbf{j}$



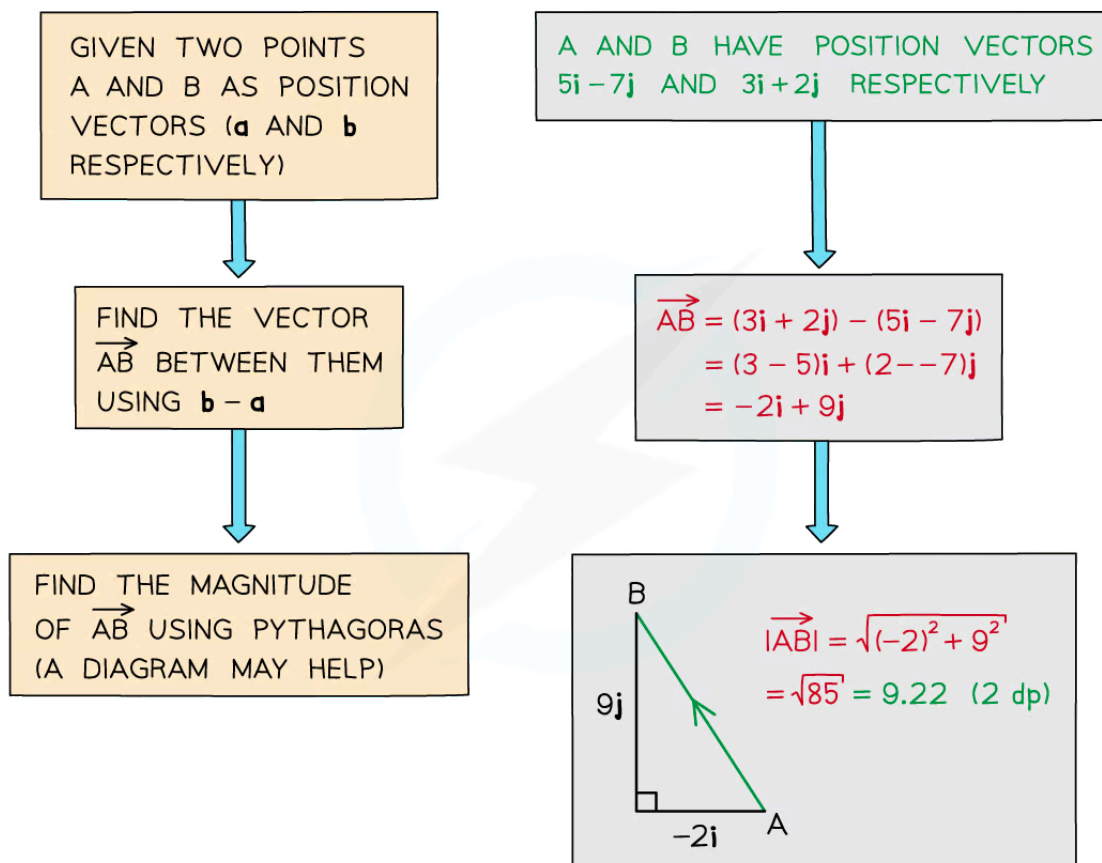
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How do I find the distance between two points using vectors?

- The distance between two points is the **magnitude** of the vector between them



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How do I find the magnitude of a displacement vector?

- You can use **coordinate geometry** to find **magnitudes** of **displacement vectors** from A to B
 - From the **position vectors** of A and B you know their coordinates

- If $\mathbf{a} = \vec{OA} = x_1\mathbf{i} + y_1\mathbf{j} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$, then point A has coordinates (x_1, y_1)

- If $\mathbf{b} = \vec{OB} = x_2\mathbf{i} + y_2\mathbf{j} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$, then point B has coordinates (x_2, y_2)



Your notes

- The **distance** between two points is given by $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
 - So $|\vec{AB}| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
- For example, if points A and B have position vectors $5\mathbf{i} + 3\mathbf{j}$ and $3\mathbf{i} - 6\mathbf{j}$ respectively
 - then $|\vec{AB}| = \sqrt{(5 - 3)^2 + (3 - (-6))^2} = \sqrt{85} = 9.22 \text{ (3 s.f.)}$
- Alternatively, you could find $|\vec{AB}|$ by
 - first using $\vec{AB} = -\vec{OA} + \vec{OB}$ to find $\vec{AB}$ in **vector form**
 - and then calculating its **magnitude** directly
 - See the Worked Example below



Examiner Tips and Tricks

- Remember if asked for a position vector, you must find the vector all the way from the origin
- Diagrams can help, if there isn't one, draw one

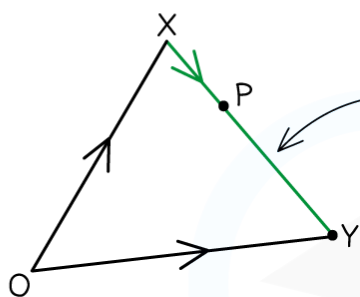


Worked Example

? The points X and Y have position vectors $3\mathbf{i} + 5\mathbf{j}$ and $6\mathbf{i} - 4\mathbf{j}$ respectively.

Point P is on the line XY such that $XP : PY = 1 : 2$.

Determine the position vector of P.



START BY DRAWING A DIAGRAM TO HELP

FIND VECTORS FOR ANY UNKNOWN PARTS OF THE DIAGRAM

$$\begin{aligned}\vec{XY} &= \vec{OY} - \vec{OX} = (6\mathbf{i} - 4\mathbf{j}) - (3\mathbf{i} + 5\mathbf{j}) \\ &= (6 - 3)\mathbf{i} + (-4 - 5)\mathbf{j} \\ &= 3\mathbf{i} - 9\mathbf{j}\end{aligned}$$

$$\vec{XY} = 3\mathbf{i} - 9\mathbf{j}$$

$XP:PY = 1:2$
MEANS P IS $\frac{1}{3}$ OF
THE WAY ALONG $\vec{XY}$

$$\begin{aligned}\vec{XP} &= \frac{1}{3}(3\mathbf{i} - 9\mathbf{j}) \\ &= \mathbf{i} - 3\mathbf{j}\end{aligned}$$

WE ARE ASKED FOR POSITION VECTOR OF P SO MUST START AT THE ORIGIN

$$\begin{aligned}\vec{OP} &= \vec{OX} + \vec{XP} = (3\mathbf{i} + 5\mathbf{j}) + (\mathbf{i} - 3\mathbf{j}) \\ &= (3 + 1)\mathbf{i} + (5 - 3)\mathbf{j} \\ &= 4\mathbf{i} + 2\mathbf{j}\end{aligned}$$



Your notes

Vector Addition

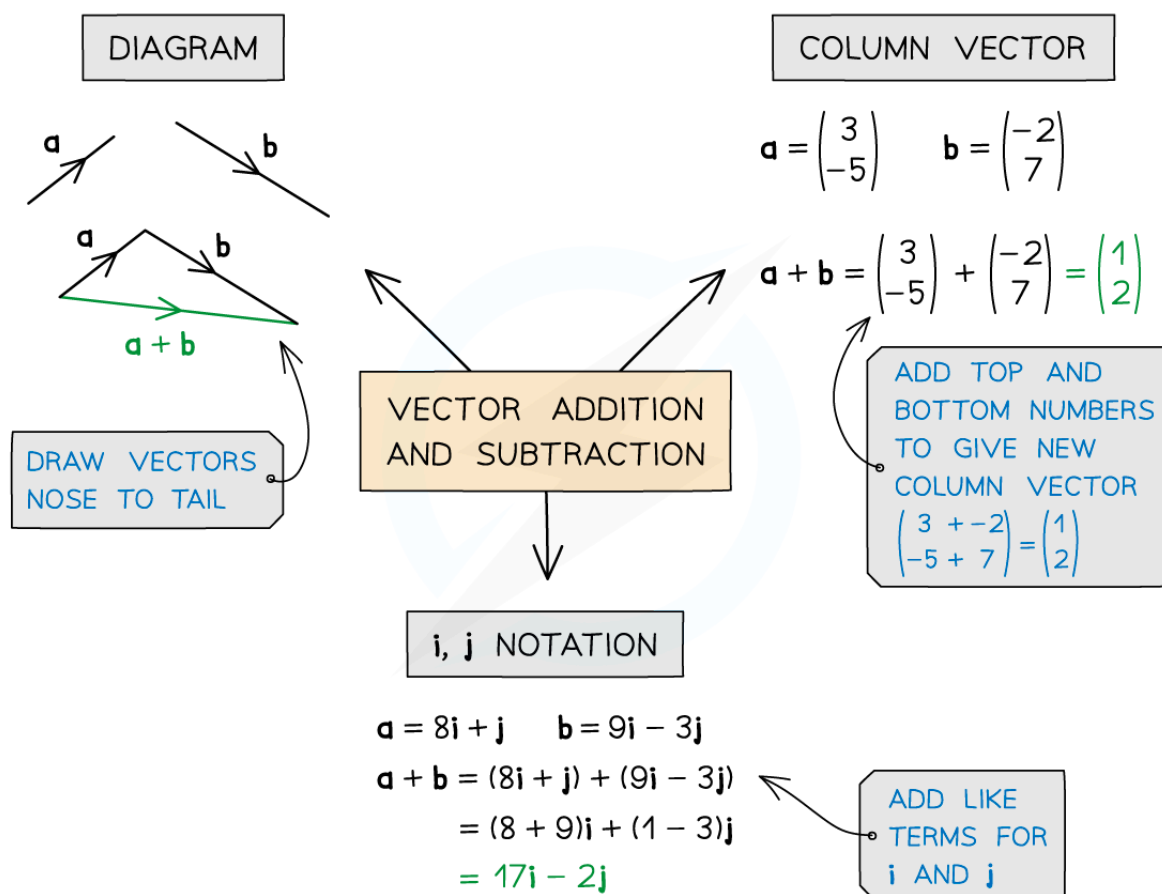
Vector Addition

What is vector addition?

- **Adding** vectors together lets us describes the **movement between two points**
- To **add** or **subtract** vectors numerically simply add or subtract each of the corresponding components
- In **column vector** notation just add the top, middle and bottom parts together
 - For example: $\begin{pmatrix} 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$
- In **base vector** notation add each of the **i** and **j** components together separately
 - For example: $(2\mathbf{i} + \mathbf{j}) - (\mathbf{i} + 4\mathbf{j}) = (\mathbf{i} - 3\mathbf{j})$



Your notes

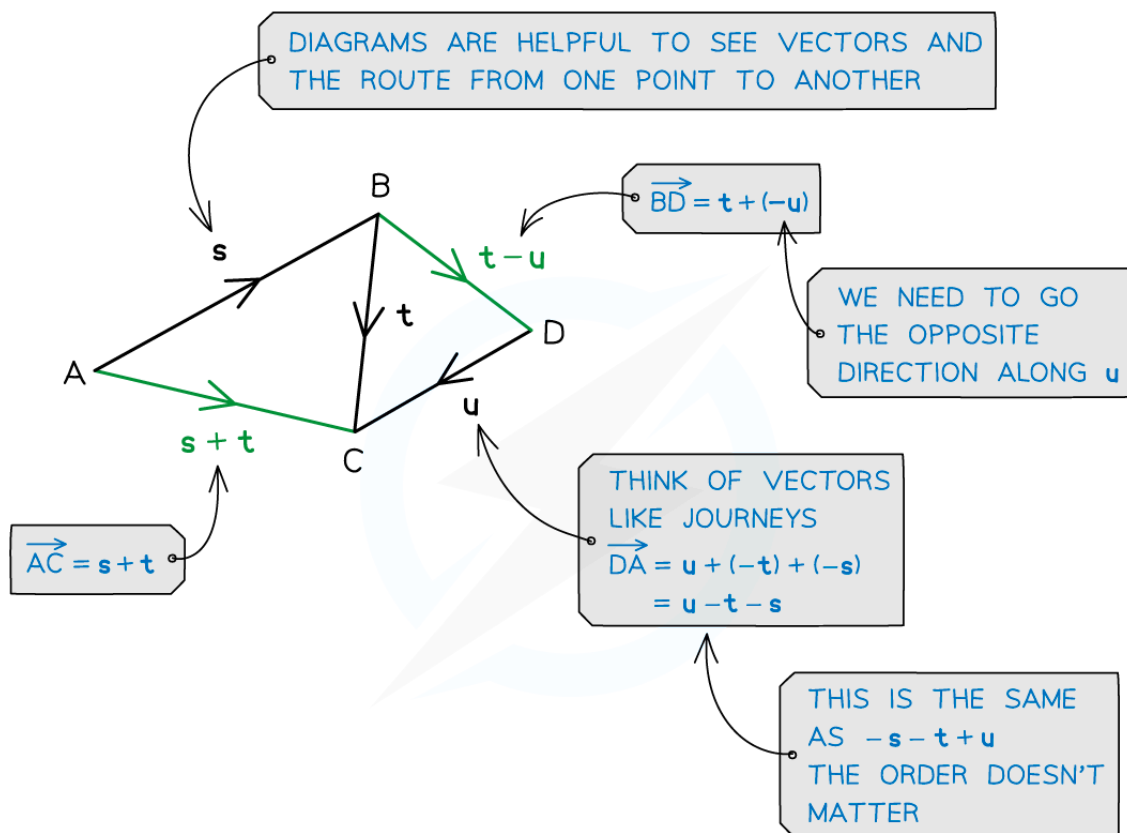


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- Adding vectors creates a **single** vector which is called the **resultant vector**
 - The **resultant** vector will be the shortest route from the start of the first vector to the end of the second
- **Subtracting** a vector is the same as **adding a negative** vector
- Adding the vectors **PQ** and **QP** gives the zero vector, denoted by a bold zero **0** (0 in handwriting)



Your notes



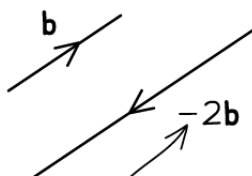
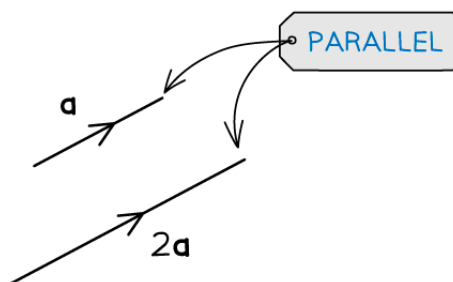
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What are scalars and parallel vectors?

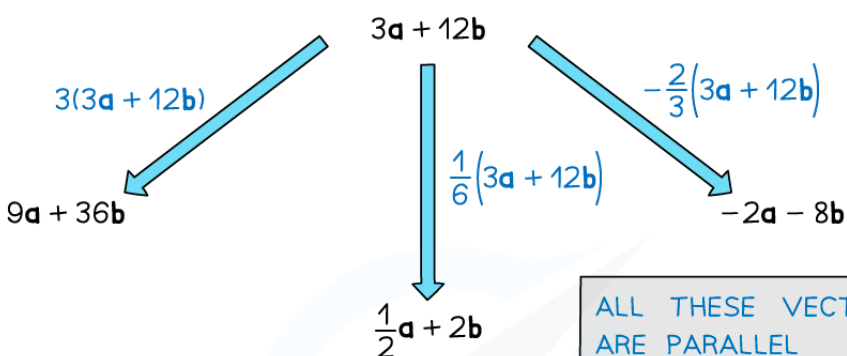
- Two vectors are **parallel** if and only if one is a **scalar multiple** of the other
 - i.e both **components** of the vector have been **multiplied** by the **same constant**
- Multiplying a vector by a **positive scalar** changes the **magnitude** (size) but **not** its **direction**
- Multiplying a vector by a **negative scalar** changes the magnitude and the **direction** would be **reversed**



Your notes



IF THE SCALAR IS NEGATIVE
THE DIRECTION WILL CHANGE



ALL THESE VECTORS
ARE PARALLEL

$$\mathbf{a} = \begin{pmatrix} 3 \\ -5 \end{pmatrix} \quad 2\mathbf{a} = 2 \times \begin{pmatrix} 3 \\ -5 \end{pmatrix} = \begin{pmatrix} 6 \\ -10 \end{pmatrix}$$

FOR COLUMN VECTORS
JUST MULTIPLY TOP
AND BOTTOM NUMBERS
BY THE SCALAR

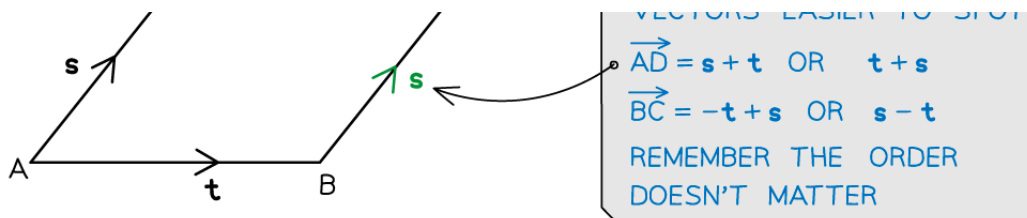
$$\mathbf{a} = 3\mathbf{i} + 7\mathbf{j} \quad 3\mathbf{a} = 3(3\mathbf{i} + 7\mathbf{j}) = 9\mathbf{i} + 21\mathbf{j}$$

FOR $\mathbf{i}$ $\mathbf{j}$ VECTORS
JUST MULTIPLY
BOTH NUMBERS
BY THE SCALAR

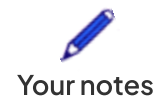
ABCD IS A PARALLELOGRAM
MEANING $\overrightarrow{AB} = \overrightarrow{CD}$ AND $\overrightarrow{AC} = \overrightarrow{BD}$



LABEL PARALLEL SIDES ON
DIAGRAMS, TO MAKE NEW
VECTORS EASIER TO SPOT



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How do I find the vector between two points?

- If, relative to the origin O , the points A and B have **position vectors**

- $\vec{OA} = \mathbf{a}$

- $\vec{OB} = \mathbf{b}$

then

- $\vec{AB} = \vec{AO} + \vec{OB} = -\vec{OA} + \vec{OB} = -\mathbf{a} + \mathbf{b} = \mathbf{b} - \mathbf{a}$

- Similarly, $\vec{BA} = \vec{BO} + \vec{OA} = \mathbf{a} - \mathbf{b}$

- This result is particularly useful when working with **position vectors** (as the 'journey' can always go via the origin)
 - but the result applies to any set of three vectors



Examiner Tips and Tricks

- Think of vectors like a journey from one place to another
 - You may have to take a detour eg. A to B might be A to O then O to B.
- Diagrams can help, so if there isn't one, draw one
 - If there are any, labelling parallel vectors will help



Worked Example



Your notes

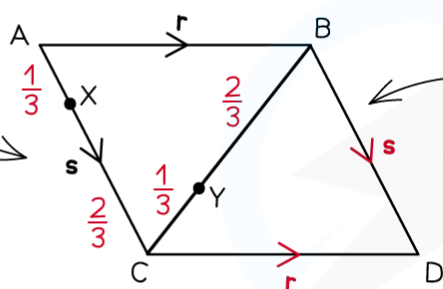
? ABCD is a parallelogram with $\vec{AB} = \mathbf{r}$ and $\vec{AC} = \mathbf{s}$.

Point X lies on $\vec{AC}$ such that $AX : XC = 1 : 2$ and

point Y lies on $\vec{CB}$ such that $CY : YB = 1 : 2$

Show that $\vec{XY}$ is parallel to $\vec{AD}$.

AX:XC = 1:2 MEANS $AX = \frac{1}{3}\mathbf{s}$
AND $XC = \frac{2}{3}\mathbf{s}$, LABEL THE
DIAGRAM TO HELP



DRAW A DIAGRAM TO HELP,
LABEL ALL PARALLEL SIDES

$$\vec{AD} = \mathbf{s} + \mathbf{r}$$

$$\vec{CB} = \mathbf{r} - \mathbf{s}$$

$$\vec{XY} = \frac{2}{3}(\mathbf{s}) + \frac{1}{3}(\mathbf{r} - \mathbf{s})$$

$$\vec{XY} = \frac{2}{3}\mathbf{s} + \frac{1}{3}\mathbf{r} - \frac{1}{3}\mathbf{s} = \frac{1}{3}\mathbf{r} + \frac{1}{3}\mathbf{s} = \frac{1}{3}(\mathbf{r} + \mathbf{s})$$

$$\vec{XY} = \frac{1}{3}\vec{AD}$$

THEY ARE SCALAR MULTIPLES
SO MUST BE PARALLEL

$$\vec{XY} = \frac{2}{3}(\vec{AC}) + \frac{1}{3}(\vec{CB})$$

THINK ABOUT THE SEPARATE
PARTS OF THE "JOURNEY"



Your notes

Problem Solving using Vectors

Problem Solving using Vectors

What problems may I be asked to solve involving vectors?

- Showing that two lines or vectors are **parallel**
 - Two vectors are parallel if they are scalar multiples of each other
 - i.e. $\mathbf{a} = k\mathbf{b}$ where k is a constant
 - See **Vector Addition**
- Finding the **midpoint** of two (position) vectors
- Showing that three points are **collinear**
 - **Collinear** describes points that lie on the same **straight line**
 - e.g. The points $(-2, -2)$, $(3, 3)$ and $(8, 8)$ all lie on the line with equation $y = x$
 - Vectors can be used to show this, and similar, results
- Results concerned with **geometric shapes**
 - Shapes with **parallel lines** are often involved
 - e.g. **parallelogram, rhombus**
 - These often include lines or vectors being split into ratios
 - e.g. The point Q lies on the line PR such that $\overrightarrow{PQ} : \overrightarrow{QR} = 3 : 1$

How do I find the midpoint of two vectors?

- If the point A has position vector $\mathbf{a}$ and the point B has position vector $\mathbf{b}$
 - the **position vector** of the **midpoint** of AB is $\frac{1}{2}(\mathbf{a} + \mathbf{b})$
- This can be derived by considering
 - $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$
 - using the result from **Vector Addition**



Your notes

- If M is the midpoint of AB then

- $\vec{AM} = \frac{1}{2} \vec{AB}$

- Therefore, the position vector of the midpoint, $\vec{OM}$ is

- $\vec{OM} = \vec{OA} + \vec{AM} = \mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a})$ $\vec{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$

How do I show three points are collinear?

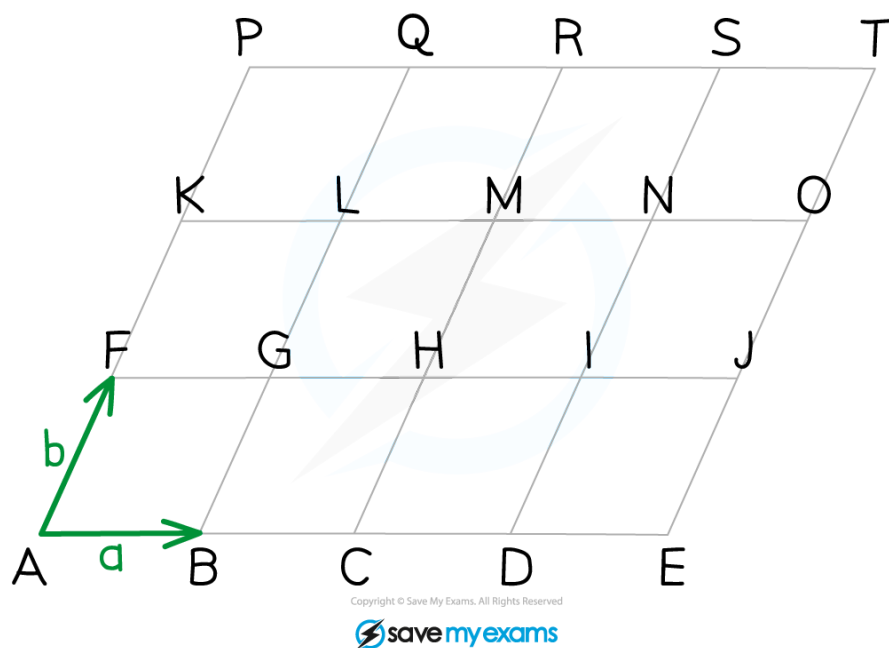
- Three points are collinear if they all lie on the **same straight line**
- There are two ways to show this for three points, A , B and C say
 - *Method 1* Show that $\vec{AB} = k\vec{AC}$ where k is a constant i.e. show that $\vec{AB}$ and $\vec{AC}$ are **scalar multiples** of each other
 - As the vectors are scalar multiples they will have the **same direction** (and so be **parallel**)
 - So as both vectors start at point A , they must be **collinear**
 - *Method 2* Show that $\vec{AB} = k\vec{BC}$ **AND** that point B lies on both the vectors $\vec{AB}$ and $\vec{BC}$
- Which method you should use will depend on the information given and how you happen to see the question

How do I solve problems involving geometric shapes?

- Problems involving **geometric shapes** involve finding paths around the shape using **known vectors**
 - there will be many other vectors in the shape that are **equal** and/or **parallel** to the known vectors
- The following grid is made up entirely of **parallelograms**, with the vectors $\mathbf{a}$ and $\mathbf{b}$ defined as marked in the diagram:



Your notes



- Note the difference between "specific" and "general" vectors
 - The vector $\overrightarrow{AB}$ in the diagram is specific and refers only to the vector starting at A and ending at B
 - However, the vector $\mathbf{a}$ is a general vector
 - any vector the **same length** as $\overrightarrow{AB}$ and parallel to it is equal to $\mathbf{a}$
 - e.g. $\overrightarrow{RS} = \mathbf{a}$
 - Vector $\mathbf{b}$ is also a general vector
 - e.g. $\overrightarrow{GL} = \mathbf{b}$
 - There will also be vectors in the diagram that are the same magnitude but have the opposite direction to $\mathbf{a}$ or $\mathbf{b}$
 - e.g. $\overrightarrow{ON} = -\mathbf{a}$, $\overrightarrow{JE} = -\mathbf{b}$
- There are also many instances of the vector addition result $\overrightarrow{FB} = \mathbf{b} - \mathbf{a}$



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- e.g. $\vec{PL} = \mathbf{b} - \mathbf{a}$
- There are many **scalar multiples** of the vectors $\mathbf{a}$ or $\mathbf{b}$
 - e.g. $\vec{FI} = 3\mathbf{a}$, $\vec{IS} = 2\mathbf{b}$, $\vec{QE} = 3(\mathbf{b} - \mathbf{a})$
- Using a combination of these it is possible to describe a vector between any two points in terms of $\mathbf{a}$ and $\mathbf{b}$



Examiner Tips and Tricks

- Diagrams are helpful in vector questions
 - If a diagram has been given, label it and add to it as you progress through a question
 - If a diagram has not been given, draw one, it does not need to be accurate!



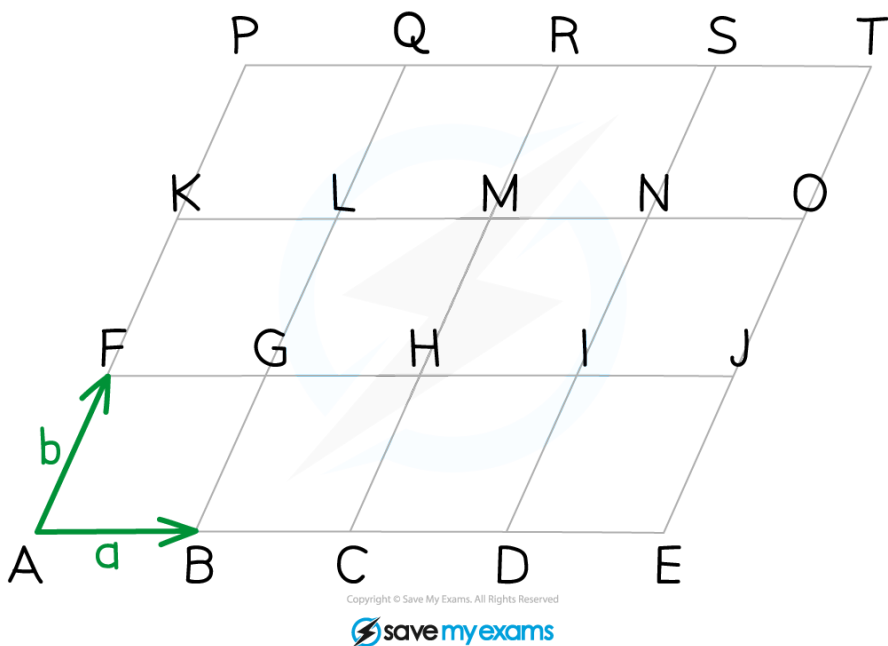
Worked Example

The following diagram consists of a grid of identical parallelograms.

Vectors $\mathbf{a}$ and $\mathbf{b}$ are defined by $\mathbf{a} = \vec{AB}$ and $\mathbf{b} = \vec{AF}$.



Your notes



Write the following vectors in terms of **a** and **b**.

a) $\overrightarrow{AE}$

To get from A to E follow vector **a** four times (to the right).

$$\begin{aligned}\overrightarrow{AE} &= \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} \\ &= \mathbf{a} + \mathbf{a} + \mathbf{a} + \mathbf{a}\end{aligned}$$

$$\overrightarrow{AE} = 4\mathbf{a}$$

b) $\overrightarrow{GT}$



Your notes

There are many ways to get from G to T . One option is to go from G to Q ($\mathbf{b}$ twice), and then from Q to T ($\mathbf{a}$ three times).

$$\begin{aligned}\vec{GT} &= \vec{GL} + \vec{LQ} + \vec{QR} + \vec{RS} + \vec{ST} \\ &= \mathbf{b} + \mathbf{b} + \mathbf{a} + \mathbf{a} + \mathbf{a}\end{aligned}$$

$$\vec{GT} = 3\mathbf{a} + 2\mathbf{b}$$

c) Point Z is such that it is midpoint of HM .

Find the vector $\vec{PZ}$.

There are many ways to get from P to Z .

One option is to go from P to R ($\mathbf{a}$ twice), and then from R to Z ($-\mathbf{b}$ one-and-a-half times).

$$\begin{aligned}\vec{PZ} &= \vec{PR} + \vec{RZ} \\ &= \mathbf{a} + \mathbf{a} - \mathbf{b} - \frac{1}{2}\mathbf{b}\end{aligned}$$

$$\vec{PZ} = 2\mathbf{a} - \frac{3}{2}\mathbf{b}$$



Your notes

Compose & Resolve Velocities

Modelling Velocities with Vectors

How are velocities modelled by vectors?

- Although introduced as being about paths and distances between points, a **vector** can also represent a **velocity**

- For example the **velocity vector** $\begin{pmatrix} 3 \\ 7 \end{pmatrix} \text{ m s}^{-1}$ (or $(3\mathbf{i} + 7\mathbf{j}) \text{ m s}^{-1}$) would represent a particle or object moving

- at 3 m s^{-1} (metres per second) in the positive x direction
- and 7 m s^{-1} in the positive y direction

- Speed** is different to **velocity**

- Speed** is a **scalar** quantity, **velocity** is a **vector** quantity
- Velocity has **direction**, as well as **magnitude**
- Speed is the magnitude of velocity

Therefore, speed can be found from velocity by finding its magnitude - i.e. using Pythagoras' theorem e.g. The speed of a particle travelling with velocity $\begin{pmatrix} 6 \\ 8 \end{pmatrix} \text{ m s}^{-1}$ is

$$\sqrt{6^2 + 8^2} = 10 \text{ m s}^{-1}$$

- An object travelling with velocity $\begin{pmatrix} -6 \\ -8 \end{pmatrix} \text{ m s}^{-1}$ would have the same speed, 10 m s^{-1} but be moving in the opposite direction

How do I find the position of a particle from a velocity vector?

- In general, problems refer to a particle's **position vector**, $(x\mathbf{i} + y\mathbf{j}) \text{ m}$ (metres, from the origin) at **time** t seconds after its motion has started
 - This position vector is often called **$\mathbf{r}$**
- If the particle moves with (constant) velocity $\mathbf{v} \text{ m s}^{-1}$, after t seconds its position vector will be



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- $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$ where $\mathbf{r}_0$ is the **initial position** of the particle (i.e. its position at the start of the motion)
- For example if a particle starts at the point with coordinates $(2, -3)$ and moves with (constant) velocity $(4\mathbf{i} + 7\mathbf{j}) \text{ m s}^{-1}$ after 10 seconds its position will be

$$\mathbf{r} = (2\mathbf{i} - 3\mathbf{j}) + 10(4\mathbf{i} + 7\mathbf{j})$$

$$\mathbf{r} = ((2 + 40)\mathbf{i} + (-3 + 70)\mathbf{j})$$

$$\mathbf{r} = (42\mathbf{i} + 67\mathbf{j}) \text{ m}$$

How do I solve problems involving velocities and vectors?

- Solving problems involving velocity may involve using a variety of the skills covered in the vectors section
 - A **resultant velocity** may be comprised of two (or more) velocities
 - e.g. the velocity of a javelin will be influenced by the athlete's ability and the wind speed/direction
 - a **resultant velocity** is found by **adding velocity vectors**
- Problems may be phrased to distinguish the difference between **speed** and **velocity**
 - Speed is the **magnitude** of velocity (use Pythagoras' theorem)
- Problems may use **position vectors**
 - The **initial position** ($\mathbf{r}_0$) is **not** necessarily the **origin**
 - $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$
- There could be **two particles** to deal with in a problem
 - be clear about which particle has which velocity, position, etc
 - If two particles **collide** at time t seconds, then (at time t) their **position vectors** will be **equal**
 - Two vectors are equal if their components are equal



Examiner Tips and Tricks

- Vector diagrams drawn previously to show paths and distances can still be used to visualise velocities



Your notes

- So use any given diagram, and if there isn't one, draw one!
- Read questions carefully – a common mistake is to give a final answer as a position vector when the question has asked for coordinates, or vice versa
- e.g. A particle with position vector $(3\mathbf{i} - 2\mathbf{j})$ has coordinates $(3, -2)$



Worked Example

A boat leaves a harbour (the origin) and sails with a (constant) resultant velocity comprising of the velocity produced by the boat's engine, $(\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-1}$ and the velocity produced by the water current, $(1.5\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-1}$.

a) Find the resultant velocity of the boat.

The resultant velocity will be the sum of the two velocities.

$$\begin{aligned}\mathbf{v} &= (\mathbf{i} + 2\mathbf{j}) + (1.5\mathbf{i} + 2\mathbf{j}) \\ \mathbf{v} &= (2.5\mathbf{i} + 4\mathbf{j})\end{aligned}$$

The resultant velocity of the boat is $(2.5\mathbf{i} + 4\mathbf{j}) \text{ m s}^{-1}$.

A second boat has position vector $(27\mathbf{i} + 6\mathbf{j}) \text{ m}$ at the same time as when the first boat leaves the harbour.

The second boat sails with (constant) resultant velocity $(-2\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$.

b) Without intervention, the two boats will collide at time t seconds.

i) Find the value of t .

ii) Find the coordinates of the point at which the boats will collide.

i) The position of a particle at time t is $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$ where $\mathbf{r}_0$ is the initial position and $\mathbf{v}$ is its (resultant) velocity.

The two boats will be in the same position when they collide.

$$\therefore (0\mathbf{i} + 0\mathbf{j}) + t(2.5\mathbf{i} + 4\mathbf{j}) = (27\mathbf{i} + 6\mathbf{j}) + t(-2\mathbf{i} + 3\mathbf{j})$$

Equating $\mathbf{i}$ (or $\mathbf{j}$) components gives an equation in t .



Your notes

$$\begin{array}{lcl} 2.5t = 27 - 2t & & \\ 4.5t = 27 & \text{or} & 4t = 6 + 3t \\ t = 6 & & t = 6 \end{array}$$

The boats collide at $t = 6$ seconds.ii) Find the position of either boat at $t = 6$ seconds.

$$\mathbf{r} = (0\mathbf{i} + 0\mathbf{j}) + 6(2.5\mathbf{i} + 4\mathbf{j})$$

$$\mathbf{r} = (15\mathbf{i} + 24\mathbf{j})$$

The question asks for coordinates (rather than a position vector).

The boats will collide at the point $(15, 24)$.